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# Leitax (A)

Brian J. McMillan apologized to Kevin C. Fowler, senior director of planning and fulfillment, as he reclined in his chair at the conference table. It was mid-January 2005 and McMillan, who had opted for back surgery during the holiday break, was frustrated that his recuperation was not proceeding as quickly as his surgeon had promised. McMillan, head of the Demand Management Services (DMS) group, and Fowler were discussing possible changes to Leitax's consensus forecasting process, a system at the heart of the company's 18-month-old revamp of its supply and demand planning processes. Headquartered in Memphis, Tennessee, Leitax was the seventh-largest player in the digital camera business.

The consensus forecasting process, as the name suggested, culminated in a single forecast (for each Leitax product) that was agreed to by all interested functions: finance, sales, product planning and strategy, and DMS. The supply-planning group used the forecast to drive Leitax's supply chain, and the finance group used it to set and monitor progress toward profitability and growth targets. The changes introduced to the supply and demand planning processes over the previous 18 months were thought to have yielded improvements in Leitax's operating performance, which included a 50% increase in forecast accuracy,[1] virtually no excess or obsolescence costs,[2] a 37% decrease in inventory levels, and an increase in inventory turns from 1.8 to 5.3 in 2004.

Two events were forcing Fowler and McMillan to revisit the forecasting process. During 2004, the consensus forecasting process had overestimated the demand for a new product family, the SF-6000, an 8-megapixel camera for the high-end consumer market. The process had also underestimated the cannibalization of the ShootXL by another newly introduced product, the Optix-R. Three-month-ahead forecast accuracy had dropped to 30% and 45% for the SF-6000 and ShootXL, resulting in write-offs of 1% and 3% of lifetime volume materials cost, respectively, on both products. Furthermore, the scheduled end-of-life date for the SF-6000 had to be extended by six months. Fowler and McMillan believed that further changes to the process would reduce these types of errors.

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[1] Forecast accuracy = 1 minus the ratio of the absolute deviation of sales from forecast to the forecast = 1 – ABS(forecast – sales)/forecast.

[2] "Obsolescence costs" were inventory charges to remove excess units from the supply chain once a product family had been discontinued.

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Professors Noel Watson and Rogelio Oliva (Texas A&M University) and Research Associate Laura Winig, Global Research Group, prepared this case. All names along with other details of the case have been disguised. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Fowler and McMillan's main concern about potential changes to the forecasting process was that the changes not interfere with the principles that underpinned the process they had put in place. These included efficient and timely sharing of information and assumptions, and respectful consideration of all points of view in reaching the consensus that would drive the organization's financial and operations planning. Fowler and McMillan needed to set the agenda for the upcoming consensus meetings, one in four days and the other in a month. The forecast to drive the FY'06 financial plan, which would also be shared with the board of directors and Wall Street, would be developed at the second meeting.

## Digital Camera Market

The digital camera was considered a breakthrough product in the photographic industry. While conventional film cameras depended entirely on chemical and mechanical processes, digital cameras had built-in computers to record images electronically, without the need for chemical developing. Like a conventional camera, a digital camera had a lens that was used to capture the light that bounced off a subject. However, the digital camera focused the light onto an image sensor called a charge-coupled device (CCD) which converted the light into tiny colored dots called pixels. The amount of detail that a camera could capture was called the resolution; the more pixels a camera had, the more detail it could capture and the higher the resolution.

By 1998, prices were falling and for less than $300 a quality camera could be purchased using 640x480-pixels (307,000 pixels)—a resolution appropriate for sharing images on an electronic monitor or computer screen, but too grainy to yield paper prints. While 640x480 pixels was pretty much the norm in 1997, by 1998 the bar had risen to 1024x768 pixels, with some consumer digital cameras having CCDs of up to 1280x1024 pixels, yielding good paper prints in 4"x5" or 4"x6" sizes. At that time consumers could purchase cameras that produced high-resolution, 8"x10" paper images using 1600x1200 pixels (2 megapixels)[3] or higher pixel count cameras. Simultaneously, digital camera prices began to fall with the introduction of CMOS (complementary metal oxide semiconductor) image sensors. CMOS sensors were much less expensive to manufacture and consumed less power than CCD censors, but offered consumers lower resolution, lower sensitivity, and lower quality. Nevertheless, manufacturers began to install CMOS sensors into their lower-end products and were consequently able to price cameras as low as $199.

Most digital cameras had an LCD screen, which allowed users to view their pictures immediately. Early generations of digital cameras had fixed storage inside the camera; users needed to connect their cameras via cable to a computer to transfer the images. By 2000, most digital cameras used removable storage devices and data compression software to make the files as small as possible.

By 2003, digital cameras were outselling film cameras, and many included features not found in film cameras such as the ability to record video and audio. Indeed, Kodak announced in January 2004 that it would halt manufacturing of reloadable (as opposed to disposable) film cameras in 2005. The market, however, was showing signs of saturation as the worldwide growth rate for digital cameras began to plummet. For 2005, the worldwide sales growth rate was estimated to be 21%; for 2006, 5.2%.[4] The estimated forecasts peaked in December 2006 when the penetration rate for digital

[3] 1 megapixel is approximately one million pixels.

[4] Susan Berfield, "The Big Picture," *BusinessWeek*, June 13, 2005, available from http://www.businessweek.com/@@1qP8q4YQn2c5FAEA/magazine/content/05_24/c3937013_mz003.htm, accessed July 8, 2005.

cameras in the U.S. was expected to be 63%.[5] The growth rate in 2007 and beyond was expected to be negative.

## Leitax

Newplex was a $22 billion print and imaging product company founded in the late 1970s. The company owed much of its early success to a line of computer printers; by the 1990s Newplex manufactured projectors, inkjet and laser printers, scanners, and other digital imaging products. Newplex sought to leverage its imaging technology investments and targeted the emerging digital camera market. After several fits and starts, the company released a 1-megapixel consumer camera. A year later, the consumer digital market was growing rapidly and the company unveiled the first in a series of pocket-sized digital cameras. In 1997, Newplex spun off its digital camera division as a separate company, Leitax. In that same year, Leitax introduced its digital LX280 Zoom camera—the first point-and-shoot megapixel consumer digital camera priced under $1,000.

Leveraging Newplex's relationship with members of its distribution network, by 2002 Leitax had rapidly become a global seller with moderate success, generating $423 million in revenue and a net income loss of $7.5 million in 2002.

By the fall of 2004, the U.S. digital camera market was $8 billion, up from $5.7 billion in 2003.[6] Digital camera shipments in the U.S. rose to 22 million units—a 36% increase over 2003.[7] (See **Exhibit 1** for market share data by manufacturer.) Leitax sold its cameras primarily through such retailers as Best Buy, CompUSA, and Circuit City in North America; Carrefour in Latin America; and others in Europe and Asia.

Organized in three geographical regions—the Americas; Europe, the Middle East, and Africa (EMEA); and Asia Pacific (APAC)—Leitax employed, in addition to its respective directors, separate sales directors for Latin America and Canada. (**Exhibit 2** provides summary information for the regions.) Distribution centers (DCs) were located in Little Rock, Arkansas; Amsterdam, Holland; and Hong Kong, China. The Little Rock DC sold directly to resellers; the other DCs sold to wholesalers who then shipped to resellers. Production was handled by two contract manufacturing plants located in China, and one each in Mexico and Chile. Contract manufacturers made either retail-ready or viewer units. Retail-ready items were packaged units ready to be shipped through the DCs to resellers, and viewer-ready units were "packed out" by DCs and made retail-ready.

On average, Leitax maintained eight camera models in its product portfolio offering a broad optical zoom range, megapixel resolution, and internal memory capacity. Each model could have multiple SKUs based on such variables as bundled accessories and geographic and promotional packages. (**Exhibit 3** lists models in Leitax's portfolio during the fall of 2004.) Digital camera product life averaged about 17–22 months and was getting shorter. High-end, feature-packed products tended to have the shortest product lives. (**Exhibit 4** shows samples of digital camera product life cycles.)

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[5] "Digital cameras—whereto?" Softpedia, March 21, 2005 available from Softpedia .com, http://news.softpedia.com/news/Digital-cameras-whereto-709.shtml accessed July 8, 2005.

[6] Ben Dobbin, "Kodak Gains on Sony in U.S. Digital Camera Market," Associated Press, November 20, 2004, available from Post-Gazette.com, http://post-gazette.com/pg/04325/414626.stm, accessed June 1, 2005.

[7] "Kodak Overtakes Sony in U.S. Digital Camera Shipments," Japan Computer Industry Scan, February 7, 2005, available from Factiva, http://www.factiva.com, accessed June 1, 2005.

## Leitax in Crisis and Supply Chain Management Response

By the end of 2002, Leitax had suffered through poor planning of three camera models: that year, the launch of one camera was delayed, another outsold its inventory, and a third reported sluggish sales. To compensate, Leitax extended the life of an existing model and made a mad scramble to find product and customers for the extended product. It was easier to find product than customers, as some customers preferred to wait for the delayed camera. One executive estimated the cost of this delay—including lost sales and the opportunity cost of inventory, warehousing space, and capital that had been set aside for the launch (which had to be written off)—to be $19.5 million. The lost sales on the second camera were estimated to be about $4.5 million in gross margins, while excess and obsolescence costs on the third model were estimated to be $2.5 million.

Executives at Newplex were concerned. The young company had failed to ensure an adequate supply of product when the market was still showing a healthy growth rate. The expected reduction in growth rate also meant that, going forward, operational efficiency was even more vital for survival. The young company had obviously not learned this aspect of the business since its inception. The high growth rate may have masked some of these inefficiencies, or the development of more efficient processes had taken a back seat to entry and growth in the market.

The inventory write-offs were followed by major changes during late 2002 as Newplex appointed a new CEO and several new vice presidents. The newly hired senior vice president for Global Operations, Luis Cruz, had more than 15 years of supply chain experience in the consumer electronics industry and was an avid amateur photographer. He quickly realized that if Leitax were to survive, the entire supply chain operation would need to be redesigned. Cruz believed that his supply chain re-engineering objectives needed to include the reduction of the cost and risks of production, as well as the cost associated with procurement, lead times, quality, and technical support. Development and implementation of this redesign was project-based, which necessitated, Cruz believed, the participation of a special type of individual. He explained:

> I have discovered that when it comes to managing demand, changing the organization for the better is really all about people, assuming that the organization has the rudimentary supply chain methods in place. No doubt I can go to the pool of top MBA grads in operations management and hire very good technical skills, people who can blend statistics, forecasting methodologies, and IT knowledge with factory skills. But putting on top of that the leadership abilities required to build an organization, know-how to influence senior management and to change the processes reduces that pool of talent significantly. These aren't people who grow on trees; they just don't exist in large numbers.

One such individual was Fowler, who was recruited in December 2002 from a rival company. Cruz valued both Fowler's technical abilities and his ability to effect organizational change. In particular, Cruz wanted Fowler to define and manage what became known officially as the Redesign Project. Initiated in April 2003, the Redesign Project had four objectives: (1) to reduce inventory levels across the supply chain; (2) to increase velocity and accuracy of planning information throughout the supply chain; (3) to increase supply upside and downside flexibility; and (4) to improve on-time performance to customers. Separate teams were to focus on the supply/demand planning process and systems; distribution and channel inventory reduction; and supplier and contract manufacturer inventory liability reduction.

Fowler was directly responsible not only for defining and managing the Redesign Project, but also for the supply/demand planning and distribution and channel inventory teams. In May 2003, he was promoted from the head of the new DMS (Demand Management Services) group to senior

director of planning and fulfillment. McMillan, who had been hired three months after Fowler, was made head of Demand Management Services.

## Need for Supply/Demand Planning Processes and Systems

Before 2003, the demand planning process had been ill-defined and there had been no DMS group. For new product introductions and midlife product replenishment, the sales group initially made forecasts, which were then informally distributed to the operations and finance groups—sometimes through discussions in the hallways. The intention was that the operations group would use these forecasts as guides for communicating build or cancel requests to the supply chain. The finance group would, in turn, use the forecasts as guides for financial planning and monitoring. The forecasts from sales, however, were often mistrusted or second-guessed when they crossed into other functional areas. For instance, the sales group thought that the finance group exerted too much pressure on forecasts because, if the forecasts did not allow the company to meet its financial plans, finance would urge sales to increase its forecasts to meet the new targets. The operations group frequently created its own forecasts when it felt it could not trust the forecasts from sales, since inventory shortages or excess, or issues with suppliers that were directly the result of poor forecasts, were still primarily the responsibility of the operations group. Even marketing, when considering the effects of possible promotions, would devise its own forecasts as an alternative to the sales forecasts when they suspected a large enough discrepancy between the two.

In addition to lack of consensus on forecasts, the data most relevant to forecasting was usually inaccurate, incomplete, or unavailable. For example: for products at the end of their life cycle, it was quite common for end-of-life (EOL) dates to be extended and production extended or restarted as a result of deals the sales group was able to arrange with particular resellers. Further compounding the problems, the firm had no system for monitoring demand planning or for creating appropriate goals so improvement could be managed.

When Fowler was hired in December 2002, his initial intention to develop the demand forecasting group with a heavy use of statistical and simulation models. But he realized that demand planning required the involvement of different functional groups whose buy-in was critical. This realization gave Fowler the idea for the consensus forecasting process that he had helped develop. He elaborated:

> Looking back, I think I came to this job with probably a bit too much ego thinking that in three months I would have this thing nailed and have developed a new forecasting process and approach. My eyes were soon opened as I progressed through existing processes during the first four months. The truth was that the root of the problem was not a "classic" forecasting issue in that it was not about getting another perfect data stream; it really didn't matter. We already had a number of forecasts that functional areas were utilizing. From finance forecasting to financial plans that met their expectations "start with a revenue number and then lets see what kind of products can build to the number," passing through the sales and product planning forecasts, each with its own biases, to our own operations forecast that ignored any market information and just looked at past builds as the most stable and desirable forecast. But there was no tie, no formal sales and operations planning process. There was no getting together to discuss "what are you guys building vs. what do you want vs. what is the financial target."

Fowler and McMillan decided on two methodological changes that would help to standardize and improve the accuracy of forecasts across the organization. Traditionally the sales directors had

forecasted expected sales from Leitax's DCs into their resellers. These sales were referred to as "sell-in." Sell-in numbers tended to be a distorted signal of demand, as the sales force had an incentive to influence sell-in in the short-term, and different retailers had time-varying appetites for inventory of products. By switching the focus of the forecasting process from sell-in to sell-through—that is, the quantities sold from the resellers in North America or shipped from the resellers' DCs in other geographic sales regions—Fowler and McMillan thought they could obtain a clearer signal of market demand.

A second change they introduced to forecasting practice was to ignore capacity constraints in estimating demand. It was common at the time for sales directors to be affected by perceptions of present and future supply chain capacity when making forecasts, which resulted in a subtle form of self-fulfilling prophecy: even if manufacturing capacity became available in the future, the deflated forecasts would have positioned lower quantities of raw materials and components, which in turn limited production output.

Before 2003, the master production schedule (MPS) had been generated monthly, but not consistently so. Lack of accuracy in the forecasting process resulted in dramatic variations between editions of the MPS, and suppliers would sometimes commit to production without due diligence to determine whether they had the required capacity or materials. The Redesign Project implemented a decision support system, and more frequent (twice a month) and consistent publication of the MPS. This, together with the improvement of the demand planning process, would enable a more accurate MPS to provide clearer signals to Leitax's suppliers about short- and long-term capacity requirements.

As part of the process redesign efforts, Fowler and McMillan made a substantial investment in information technology, including SAP's Advanced Planning Optimizer (APO) suite with the following components: Demand Planning/Demand Forecasting (DP/DF), Supplier Network Planning (SNP), and Global Available to Promise (GATP). DP/DF—which was used to support the improvement of demand planning—provided forecasting capabilities such as multiple forecast collection and integration, and short- and long-term forecasting with programmable and prepackaged models. SNP took the weekly forecasts over the planning period and generated initial production requests (an initial MPS) based on available capacity of input suppliers. Finally, the GATP—used to improve the customer commitment process—gave staff a real-time, user-friendly view of Leitax's inventory so they could make accurate commitments on customer orders.

## Demand Planning: Consensus Forecasting

### *Implementation*

Having realized that buy-in from (and involvement of) the different functional groups was fundamental to changing the demand planning process, Fowler started generating functional support. He first offered assurances that, with regard to the proposed new forecasting process, the voices and perspectives of those in the different groups would be heard. Fowler took the time to acquaint each group with the company's forecast and inventory performance to date, and was quick to point out that the problem belonged not to a particular group but was collective. Instead of developing yet another forecast based on statistical methods, he created mechanisms that would ensure that a single forecast drove the organization.

In April 2003 the first consensus forecasting meeting was held, with regional sales directors providing sell-through and sell-in forecasts for their regions, and the Product Planning and Strategy group providing worldwide sell-through forecasts. These forecasts were presented under the watchful and often persuasive presence of the finance group. Early meetings were rocky as different groups defended their forecasts or points of view. It was not unusual for meetings to exceed half a day.

As a result of these early meetings, it became clear that changes to the process were needed. A significant source of differences between the forecasts was the range of assumptions about product price changes, and promotion schedules. Realizing that business assumptions were crucial to the overall process, the DMS group developed and refined a package of information that summarized promotional as well as competitors' product information. It also became obvious that the DP/DF component of SAP's APO Suite could not provide an interface with enough flexibility to collect the individual forecasts and manage the consensus forecasting meetings. Instead, Excel templates, which could also provide input to the DP/DF component, were created for this purpose. The DMS group made incremental changes to the forecasting process over the next 18 months.

## *Process*

By the fall of 2004, consensus forecasts were generated monthly by a group that included the VPs of marketing, product strategy, finance, and product management, as well as the sales directors, some of whom participated in the meetings via telephone conference call. The goal of the consensus forecasting process was to collect assumptions about future product demand and performance from different areas of the organization and to summarize these assumptions in a single document that would be used to drive sales, manufacturing, and financial planning. The process in itself created multiple opportunities for evaluating and revising these assumptions and their implications.

The starting point for the consensus forecasting process was the business assumptions package (BAP), which contained price plans for each SKU, intelligence about market trends and competitors' products and marketing strategies, and other information that was relevant to the industry in general. The product planning and strategy (PPS), marketing, and DMS groups helped to assess the impact of the information on future business performance that they entered into the BAP. (The BAP was an Excel document with multiple tabs for different types of information and an accompanying PowerPoint presentation.) The recommendations were carefully labeled as such and were generally made in very broad terms. The BAP typically reflected a one-year horizon. It was updated monthly and was discussed and agreed on by the forecasting group during a monthly two-hour meeting. (**Exhibit 5** provides sample contents of the BAP.) Once the information in the BAP was agreed on, three functional groups (PPS, Sales, and DMS) used it as the basis for additional forecasts.

**Top-down forecast from PPS** PPS, a four-person group that provided support in all aspects of product life cycle from launch to end-of–life, and which assessed competitive products and effects of price changes on demand, prepared a *top-down* forecast of the global expected demand for digital cameras. The PPS forecast was derived from a worldwide estimate of digital camera demand, and calculated product- and region-specific forecasts based on historical and current trends of Leitax's market share as well as the current portfolio of products being offered by Leitax and its competitors. The PPS group used research from market research groups to spot current trends and used appropriate history as precedent in assessing competitive situations and price effects.

**Bottom-up forecast from sales** The sales directors used a *bottom-up* approach to generate their forecasts. The sales organization was responsible for educating resellers about products, as well as upcoming promotions and launches. Sales directors from all regions aggregated their account

managers' and their own knowledge about channel holdings, current sales, and expected promotions to develop forecasts based on information about what was happening in the distribution channel. Sales directors were inclined to shield account managers from this forecasting responsibility, maintaining that it would be too distracting to involve them. The sales directors' bottom-up forecast was first stated as a sell-in forecast, since most incentives for the sales organization were based on commissions on sell-in to the resellers. The sell-in forecast was then translated into a sell-through forecast by assuming maintenance of a maximum level of channel inventory (inventory at downstream DCs and at resellers) for each SKU of approximately 7–8 weeks of inventory at the sell-through rate. Sales directors' compensation was based not only on top-line revenue, but also on margins and weeks of inventory in the channel, thus limiting the "upside" of an inflated forecast. The sales directors' bottom-up forecast was based on orders and retail and distribution partner feedback, and was instrumental in determining the first 13 weeks of the master production schedule.

**Sell-through forecasts from DMS** The DMS group prepared a third forecast of sell-through by region, based solely on statistical inferences from past sales. The statistical forecast was intended primarily to provide a reference point for the other two forecasts. Significant deviations from the statistical forecast required that the other forecasting groups investigate and justify their assumptions. The PPS and DMS groups' forecasts had a one-year horizon, and the sales directors' forecasts were for the nearest two quarters. All forecasts included, by region, monthly expected sales for each product type. Monthly expected sales were spread evenly over the month to yield weekly forecasts.

The three groups' forecasts were submitted in an Excel template and merged by the DP/DF module into a proposed consensus forecast using a formulaic approach devised by the DMS group that gave more weight to the sales directors' bottom-up forecasts in the short term and increased the weight of the top-down forecast in the long term. The proposed consensus forecast and the three independent forecasts were then evaluated by the forecasting group in a monthly two- to four-hour meeting. (**Exhibit 6** provides examples of the three different forecasts and the proposed consensus forecast.) The main purpose of the consensus forecasting meeting was to help all parties understand the assumptions driving the different set of forecasts, and to allow everyone to agree on a final consensus forecast based on those assumptions and their implications. Discussions in the meeting tended to focus on the nearest two quarters.

## *Consensus Forecasting Meetings*

In contrast to the first few consensus meetings which had sometimes lasted for a full day and had been characterized by heated discussions, by the fall of 2004, conversations during the meeting were cordial, lasting two to three hours. Facilitated by McMillan, participants focused on understanding the reasons behind significant differences among the functional forecasts. Once the underlying reasons for diverging forecasts were identified and discussed, the group would revise the proposed consensus forecast by open discussion. Sales' bottom-up forecasts that were slightly higher than the other two forecast types were often justified by the sales directors based on their intimate knowledge of upcoming sales deals or prospects; as a result, the higher forecasts were accepted by the group. Although the finance group did not submit a forecasting stream, its members still voiced their opinions and concerns about the forecasts given their understanding of the revenue potential at stake. Further, although the PPS group had little functional stake in the outcomes of the meetings, it tended to be vocal, objective, and unemotional about forecasts and demand planning. Topics covered at the consensus meetings included the following:

- **New product introduction** was carefully considered. Sales directors for each region considered the PPS and DMS groups requests for initial inventory for newly introduced

products when creating forecasts. Ideal launch quantities were usually 10 weeks of product at the expected sell-through rate. As a launch date drew nearer, the consensus forecasting meeting was used to report the expected inventory status at launch, revise regional requests, and seek consensus on the actual regional allocation of inventory, taking into account any expected shortfalls at launch.

- **Products to be discontinued** in the current quarter were also discussed. The proposed drop date was presented together with a sales history, predicted sales for the rest of the product life cycle, and available inventory. Consensus was then sought on how a product's end-of-life should be managed.

- For ongoing products, **sell-in and sell-through rates** and the **channel inventory** were analyzed and compared with expected sales. The finance group aggregated channel inventory and sales data from weekly reports from the resellers. Any discrepancies between inventories reported and inventories calculated from sell-in and sell-through data were resolved, and consensus sales forecasts were updated with the latest information available. Promotion or price-change decisions were also revised in light of this data.

- Finally, the consensus forecasting meeting was used to provide feedback on **forecasting performance**, emphasizing biases that functional forecasts had displayed in past meetings. For example, during recent meetings McMillan had presented evidence that the sales forecasts tended to overestimate short-term sales and underestimate long-term sales.

## *Creating the Final Consensus Forecast*

Once all details were agreed on, the Final Consensus Forecast (FCF) was sent, via another Excel worksheet, to the finance department for financial rollup. The finance department combined the consensus forecast with pricing and promotion information from the BAP to determine expected sales and profitability, thereby converting the quantities forecasted in the FCF into their revenue equivalents. Forecasted revenues were compared with the company's financial targets. If gaps were identified, the finance group would first ensure that the sales group was not underestimating the product's market potential. If revisions made at this point did not result in satisfactory financial performance, the forecasting group would return to the business assumptions and, along with the marketing department, revise the pricing and promotion strategies to meet financial goals and analyst expectations. These exercises usually occurred at the end of each quarter and could result in significant changes to forecasts. Once approved, the FCF was released and used to generate the MPS.

## *Measuring Performance: Forecast Accuracy Metric*

To assess the efficacy of the consensus forecasting process, the DMS estimated the 13-week forecasting accuracy of both the sell-in and sell-through forecasts based on the formula that reflected the fractional variance of the forecast error: FA=1-ABS(sales-forecast)/forecast. Forecast accuracy was estimated for each product line and region, and a single metric, for both sell-in and sell-through, was calculated based on the weighted (by volume) average of these individual forecasts. Thirteen weeks was selected as the appropriate lead time to assess forecast accuracy because it reflected the longest contracted lead time for components in Leitax's supply chain. As Fowler stated, "It is pointless to be great at forecasting within four weeks or eight weeks if your manufacturing lead time is greater than 13 weeks; the supply chain needs time to react."

The metric, however, was considered simplistic by some because it compared forecasts to actual sales as if the actual sales represented true demand, ignoring the effect of stock-outs. Fowler commented:

> If instead of taking actual sell-in shipments as actual demand, you consider, say, sell-in plus requested backlog that we did not fulfill, then you would start to get into a gray area. Because that backlog is cancelable within a day, some of our backlog is always phantom.[8] How much is phantom or real? No one knows that. With sell-through forecast accuracy, the question becomes, "If you could have sold-in more, would you have sold-though more?" Sell-through is not only a function of channel inventory but also depends on regions, being affected by retailer actions and even actions by individual stores. So, in either case, you get to a place where you are in such a gray area that you could almost make the data look how you want it to look. And that is where we struggled a bit. It is a valid approach to say, "Well you missed some demand, account for that," but is very hard to fix it explicitly and without looking like you are manipulating numbers. It is the right thing to do, but if you start to get into a situation that is so hard to describe that people begin to say, "What does that number mean?" then the metric loses its ability to convey information. As a performance measure, it is really a very sensitive subject; it makes everybody squirmy.
>
> I was at a seminar recently where one of the participants stood up and said, "We've given up; we are not measuring forecast accuracy anymore." Which is absolutely the wrong answer, but you can see why you get there; it's too hard to get a metric that people understand and feel measures their performance fairly. People love to hate this metric.

## *Results*

By fall 2004, improvements in forecasting accuracy at Leitax were dramatic. Three-month-ahead sell-through forecast accuracy improved from 58% in the summer of 2003 to 88% by fall 2004; sell-in forecast accuracy improved from 49% to 84% over the same time period (see **Exhibit 7**). Better forecasts, in turn, translated into significant operational improvements: Inventory turns increased to 5.3 in Q4 '04, from 3.5 the previous year, and average on-hand inventory decreased from $142 million to $101 million (see **Exhibit 8**). Excess and obsolescence costs decreased from an average of $3.1 million in 2001–2003 to practically zero in 2004.

There were lingering issues, however, with process compliance. In the DMS group's estimation, the process resulted in a channel inventory level greater than the desired 7–8 weeks more often than they would have expected. This could be explained by overly optimistic forecasts from the sales organization, especially given the way inventory shortages were handled. Under limited supply conditions, the DMS would manage inventory allocation to regions and would do so based on historical demand by region. The higher-than-expected inventory level could also be explained by the sales group's over-selling into the channel in response to their incentives. The appropriate role of the finance group within the process was still in question; there were lingering complaints by the sales group that the finance group used the consensus meetings to try to push sales to commit to higher revenues. Gap-filling exercises, although serving the purpose of constructively taking the feedback from finance and channeling it toward the business assumptions, sometimes resulted in significant changes to forecasts that seemed inappropriate.

[8] "Phantom orders" are placed by customers to increase product allocation, but without real intent of receiving it.

## Missteps

Despite the overall improvement in operational indicators for the forecasting process, two recent examples seemed to signal a need for more immediate process improvement.

### *SF-6000*

Launched in the spring of 2004, the SF-6000 was the first camera on the market to offer an 8-megapixel sensor and a 10x zoom. Priced at only $999.95, the camera was aimed at the high end of the consumer market; indeed, in its official press release, Leitax called the camera "a tool for serious photographers." The SF-6000 was broadly reviewed in the consumer and photography press, which praised its easy-to-use controls, ergonomics, and in-camera image sharpening.

When product introduction and early sales were being planned at the supply/demand process level, enthusiasm of the specialized press was reflected in the company's consensus forecasts. (See **Exhibit 9** for forecasts at the time of introduction vs. actual sales.) The changes in the company's forecasting process had provided some but not enough dampening of enthusiasm for the product. The product ended up being woefully over-forecasted and expectations were that there would be a write-off of about 1% on lifetime volume materials cost.

### *ShootXL / Optix-R*

Leitax introduced the ShootXL in April 2004. A widely recognized hit for Leitax, the ShootXL was a digital SLR that boasted a 6-megapixel CMOS sensor and image processor. With a manufacturer's suggested retail price of $799 (street price of $729), the ShootXL sat at the low end of the professional market. The model quickly became one of the best-selling digital cameras ever offered.

Leitax introduced the Optix-R, which was modeled after the ShootXL, in September 2004. The camera had the same famed 6-megapixel processor but was housed in a plastic body, used a pentamirror in the viewfinder (instead of a pentaprism), and was manufactured in China rather than Chile to reduce costs. Optix-R's manufacturer's suggested retail price was $799, but the camera could be purchased online at the deeply discounted price of $459. Even so, the Optix-R sat at the high end of the consumer market, but appealed to "prosumers" who now viewed the Optix-R as an affordable alternative to the ShootXL.

Despite warnings from the DMS group indicating possible cannibalization of the ShootXL by the Optix-R, the consensus forecast had the ShootXL continuing its impressive sales rate throughout the Optix-R introduction. (See **Exhibit 10** for sales performance of the ShootXL and sales forecast at the time of the Optix-R introduction.) The larger-than-expected cannibalization resulted in a 3% of lifetime volume materials cost obsolescence write-off for the ShootXL.

## Challenges

With the consensus forecasting process in place, Fowler and McMillan considered the future of demand management at Leitax and wondered whether it was the right time to try to introduce more sophisticated forecasting techniques to aid the demand planning process. There was sufficient evidence that the statistical forecasts, even with new product introductions, performed reasonably well. This did not surprise Fowler, since forecasting experts often found that statistical forecasts were quite robust when there was previous experience with the product type and market. Fowler,

however, realized that the responsibility for mastering and using these tools would fall to the DMS group and was concerned that further statistical forecasts might be perceived as the group forcing its decisions on the company. The company's revamp of its processes seemed to reveal a trend of increasing responsibility and control by the DMS group that could begin to work against it. Fowler was mindful, though, that there were some supporters in the company who favored less input from the sales group in forecasting.

The SF-6000 forecast miss had caused Fowler and McMillan to consider a different way of focusing the consensus meetings. In the past, meetings had largely focused on the two nearest quarters; Fowler and McMillan recommended focusing instead on lifetime quantities in meetings and letting quarter and regional breakdowns be more historically driven. Fowler and McMillan had already heard feedback on the first change when one sales director complained that without being able to discuss the details of the next two quarters in the meetings, she would not be sure what was being agreed.

Finally, the Optix-R introduction suggested that the DMS group, despite the process in place, might lack the skills and discipline to navigate the complex pricing and feature preference of customers, especially given increased overlap in product lines. Moreover, the product life of digital cameras was getting shorter, which meant more product launches and more instances of feature overlap to manage. Again, Fowler and McMillan considered whether statistical techniques could be of help.

However, before Fowler and McMillan could make a decision about this, they wanted to be sure they understood the essential elements of the consensus forecasting process. What were its weaknesses and strengths? What were the situations in which it worked well? In which it did not, and why? How could it be tweaked to maintain its fundamentals and match the operational and organizational concerns? In the beginning the situation had been easy to characterize because there was no process in place; but now, with a well-defined process in place, predicting effects of changes meant understanding the new dynamics that the process had introduced. Was tweaking enough at this point? Was a more significant change required?

Breaking for lunch, McMillan groaned as his back began the familiar ache. That the surgery had been voluntary seemed to make the pain all the more frustrating. As he hobbled off, his cell phone began to ring and he saw that it was his doctor responding to his earlier call. Hopefully, the doctor had more than just reassurances that the pain would soon go away.

**Exhibit 1** 2004 U.S. Market Share by Manufacturer for Digital Cameras

| | % Share |
|---|---|
| Kodak | 22% |
| Sony | 20% |
| Canon | 16% |
| Olympus | 10% |
| Hewlett-Packard | 8% |
| Fuji | 8% |
| Leitax | 6% |
| Other | 10% |

Source: Adapted and compiled from "Kodak Overtakes Sony in U.S. Digital Camera Shipments," Japan Computer Industry Scan, February 7, 2005, Factiva, June 1, 2005; Ben Dobbin, "Kodak Moves up in Digital-Camera Market," mercurynews.com; and IDC data.

**Exhibit 2** Breakdown of Leitax's Digital Camera Market, 2004

| Sales Region | Number of Countries | Three-Year Average Percentage of Revenue | Number of Customers 2004 |
|---|---|---|---|
| United States | 1 | 63% | 54 |
| Latin America | 10 | 5% | 16 |
| Canada | 1 | 5% | 5 |
| Europe, Middle East, and Africa (EMEA) | 28 | 21% | 67 |
| Asia-Pacific (APAC) | 10 | 6% | 25 |

Source: Company data.

**Exhibit 3** Leitax Product Line, Fall 2004

| Model | Effective Pixels | Market | Release Date | Suggested Retail | Maximum Print Size | Weight/ with Batteries | Zoom[a] | Memory[b] | Movie Recording[c] | Special Features |
|---|---|---|---|---|---|---|---|---|---|---|
| CX272 Digital | 2.0 MP | Low-end consumer | January 2004 | $99.95 | 8.5“ x 11“ | 7.6 oz. | Digital Zoom 3.0x | 6 MB Internal | 320 x 240 max 15 seconds | Compact, inexpensive |
| CX3173 Digital | 3.1 MP | Low-end consumer | Fall 2003 | $179.00 | 8.5“ x 11“ | 6.7 oz. | Digital Zoom 4.0x | 16 MB Internal | 640 x 480 max 26 sec; 320 x 240 max 52 sec | Warehouse club pack bundled with accessories |
| DX575 Digital | 4.0 MP | Consumer | Summer 2004 | $329.00 | 9“ x 12“ | 10.6 oz. | n/a | 16 MB Internal | 320 x 240, 30 fps, 10 fps, no audio | Inexpensive, no zoom |
| DX474 Zoom Digital | 4.0 MP | Consumer | Summer 2004 | $499.00 | 9“ x 12“ | 11.5 oz. | Digital Zoom 3.6x | 16 MB Internal | 640 x 480, 10 fps, max 30 sec | 4MP Zoom camera for under $500 |
| LX543 Zoom Digital | 5.0 MP | Consumer | Fall 2004 | $599.00 | 20“ X 15“ | 6.6 oz. | Digital Zoom 4x | 32 MB Memory Stick Duo | 320 x 240, 15 fps, unlimited, with audio | Low price per megapixel, memory stick included |
| Optix-R | 6.0 MP | High end consumer | Fall 2004 | $799.00 | 20“ x 30“ | 7.7 oz. | Optical Zoom 4x | 16 MB Secure Disk card | 320 x 240, 15 fps max 30 sec | Plastic body; deeply discounted street price |
| ShootXL | 6.0 MP | Professional | Spring 2004 | $799.00 | 20“ x 30“ | 7.1 oz. | Digital Zoom 4x | 16 MB, xD-Picture Card | 320 x 240, 15 fps | CMOS sensor |
| SF6000 | 8.0 MP | Professional | Spring 2004 | $999.00 | 20“ x 30“ | 1.6 lb. | Optical Zoom 10x | 32 MB xD-Picture Card | 640 x 480, 15 fps; 320 x 240, 15 fps | 10x zoom |

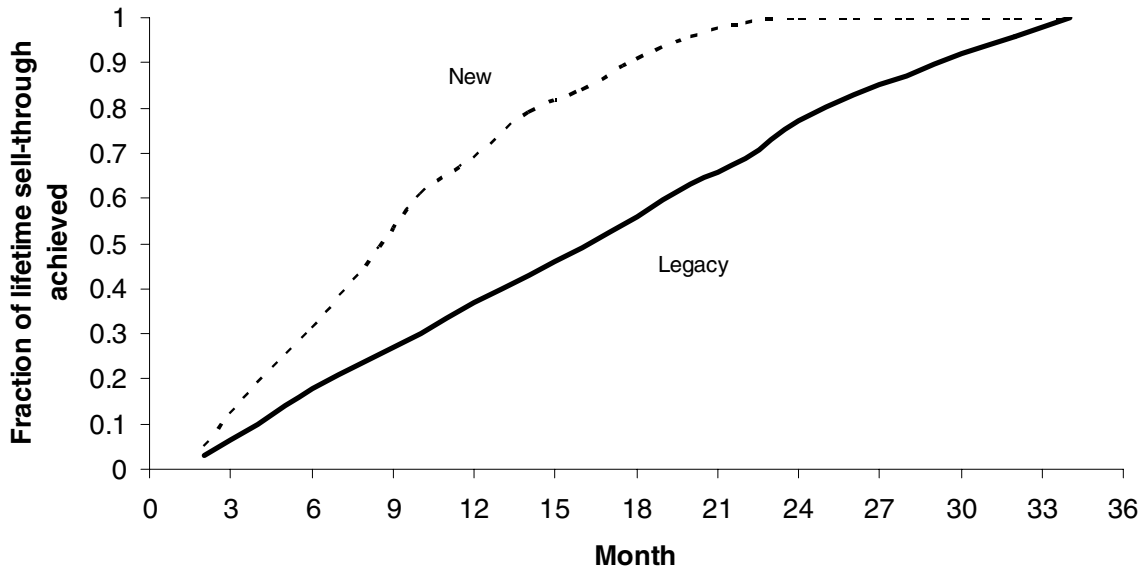
Source: Company data.

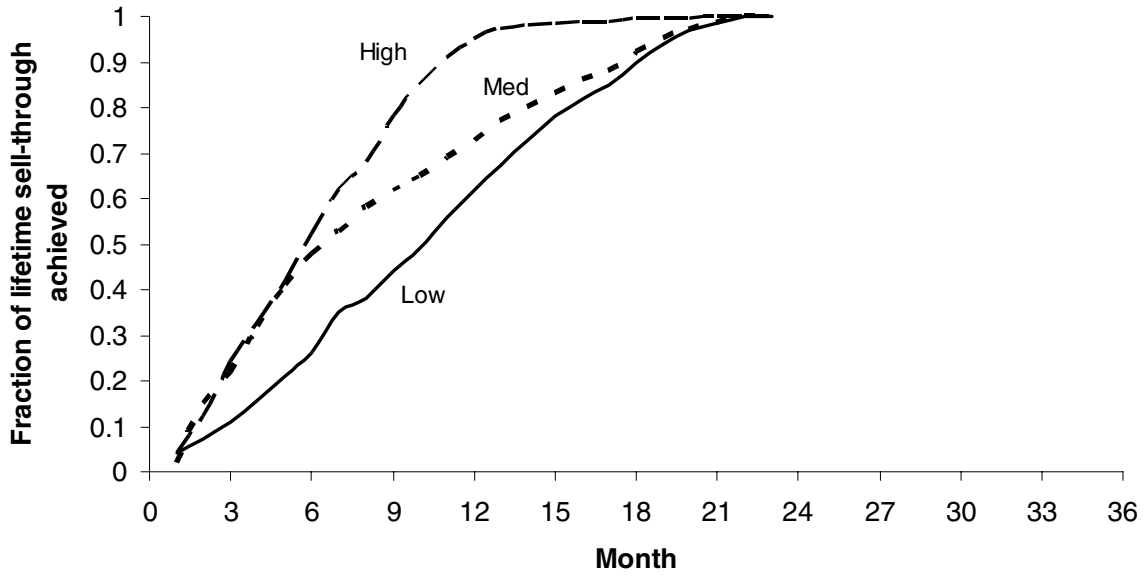
[a]Optical zoom is a true zoom: the distance of the lens to the subject is changed when the zoom is engaged. Digital zoom is unique to digital cameras and merely enlarges an area of the image. It is considered inferior to an optical zoom. Magnification measures ranged from 3 times (3.0x) to 10 times (10.0x) the size of the subject.

[b]Memory was measured in megabytes (MB); a higher number allowed for storage of more images. Memory was either fixed (internal only) or removable (cards or sticks.) Removable memory, like film, allowed users to replace storage media when capacity was reached. With internal memory, the user had to download images and remove them from the camera before more images could be captured.

[c]Movie recording options included picture resolution (320 x 240 pixel resolution was considered the minimum; 640x480 pixels was state of the art) and frames per second (fps); 30 fps would play back as normal live action, whereas fewer fps would yield a movie that appeared choppy.

**Exhibit 4** Leitax Digital Camera Product Life Cycles: New vs. Legacy, and High vs. Medium vs. Low Price Points for New Products




Source: Company data.

**Exhibit 5** Sample Contents of the Business Assumptions Package (BAP), January 14, 2005

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## Q3 FY05 Actions

Q3 Dec – Feb 2004/2005

**Announce**
**Jan 15th**
- CX272 Digital $199 (EMEA only)

**Actions & Rebates**
Dec 4th
- LX543 Zoom Digital $100 rebate continues and hard price action to $449 in EMEA 12/4

Dec 12th
- DX474 Zoom Digital $50 price action from $499 to $449

Jan 15th
- DX474 Zoom Digital $30 price action from $499 to $469 in EMEA

Feb 27th
- LX543 Zoom Digital $50 coupon (or $30 price action)
- DX575 Digital price action to $299

**End-of-Life**
Dec 1st
- CX3173
- CX3073

Feb 27th
- EMEA - C272
- APAC - C272

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## Handheld: IDC findings

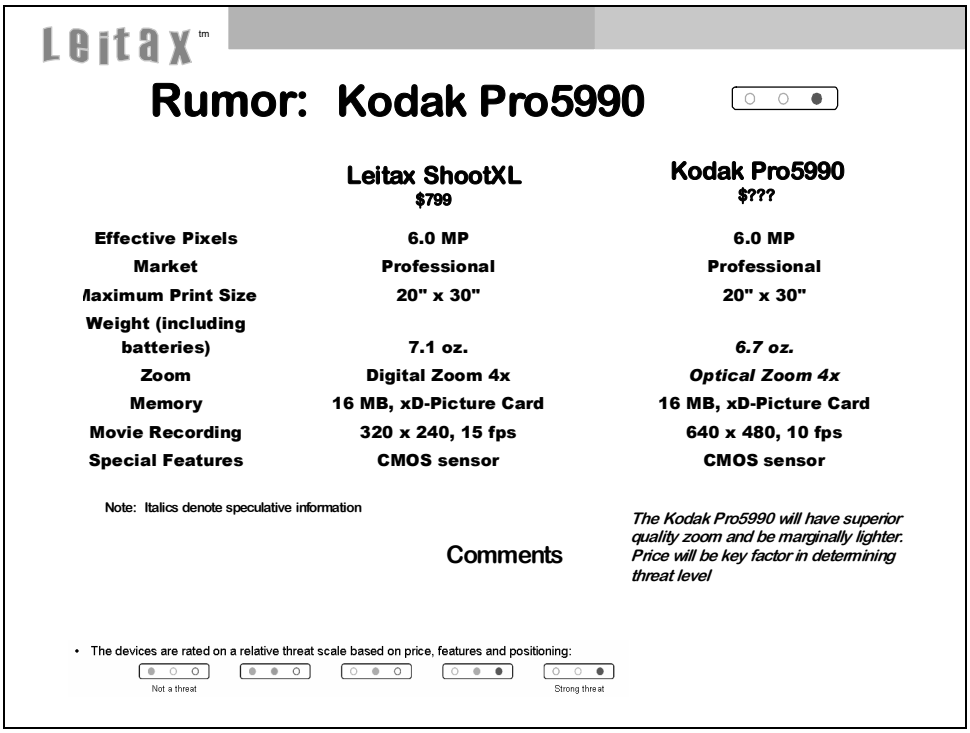
- **IDC reports growth estimate of 20.7% for 2004–2005 :**
  - **US xx.5%**
  - **Western Europe xx.7%**
  - **Japan/APAC xx.1%**
  - **Rest of World xx.3%**

## Handheld: Market Demand Overview

- **Units YTD Jan-Nov '03 vs. '04**
  - US All Channel: 14.2M vs. 19M (34%)
  - Leitax: 0.87M vs. 1.14M (31%)

IDC Dec. 2004

**Exhibit 5 (continued)**




Source: Company data.

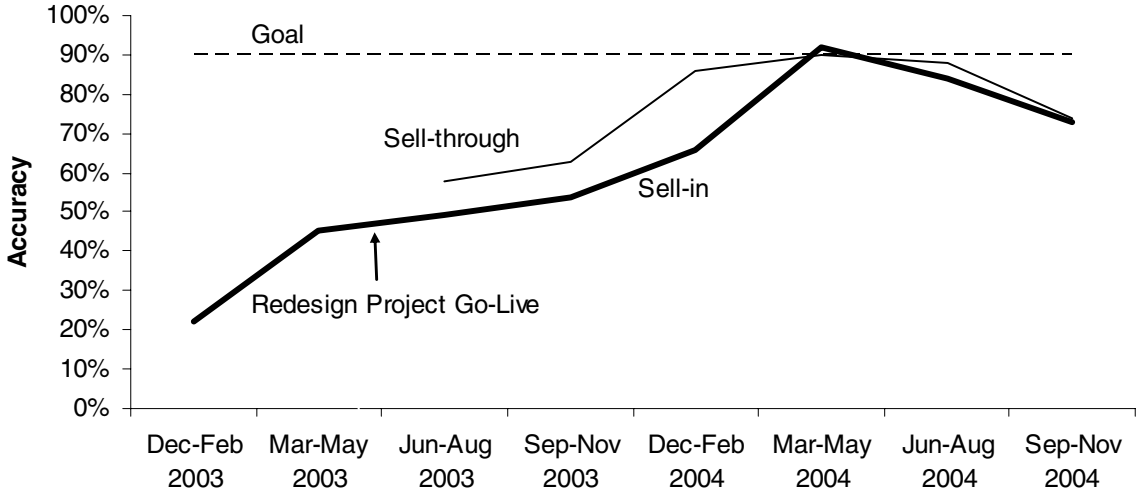
**Exhibit 6** Individual and Proposed Consensus Forecast for SF-6000 Sell-Through—July 30, 2004

| | Gross Sell-Thru Q2 | | | Q2 Total | Gross Sell-Thru Q3 | | | Q3 Total |
|---|---|---|---|---|---|---|---|---|
| | September | October | November | Sell Thru | December | January | February | Sell Thru |
| **Worldwide** | | | | | | | | |
| Sales | 6,817 | 9,510 | 10,092 | 26,418 | 10,667 | 6,800 | 9,053 | 26,520 |
| Top Down | 7,323 | 8,086 | 8,533 | 23,943 | 9,333 | 6,667 | 8,004 | 24,004 |
| Statistical | 3,779 | 4,606 | 8,601 | 16,986 | 10,008 | 7,161 | 5,010 | 22,179 |
| Proposed Consensus[a] | 7,070 | 8,798 | 9,313 | 25,180 | 10,003 | 6,876 | 7,356 | 24,234 |
| **United States** | | | | | | | | |
| Sales | 2,683 | 3,933 | 5,873 | 12,490 | 3,907 | 2,420 | 3,320 | 9,647 |
| Top Down | 3,662 | 4,043 | 4,267 | 11,971 | 4,667 | 3,333 | 4,002 | 12,002 |
| Statistical | 1,775 | 1,670 | 3,332 | 6,777 | 4,861 | 3,192 | 2,133 | 10,186 |
| Proposed Consensus[a] | 3,172 | 3,988 | 5,070 | 12,231 | 4,478 | 2,982 | 3,151 | 10,611 |
| **Canada** | | | | | | | | |
| Sales | 333 | 500 | 500 | 1,333 | 333 | 333 | 400 | 1,067 |
| Top Down | 366 | 404 | 427 | 1,197 | 467 | 333 | 400 | 1,200 |
| Statistical | 154 | 118 | 415 | 688 | 192 | 135 | 153 | 481 |
| Proposed Consensus[a] | 350 | 452 | 463 | 1,265 | 331 | 267 | 318 | 916 |
| **Latin America** | | | | | | | | |
| Sales | 250 | 583 | 250 | 1,083 | 333 | 333 | 333 | 1,000 |
| Top Down | 366 | 404 | 427 | 1,197 | 467 | 333 | 400 | 1,200 |
| Statistical | 367 | 421 | 815 | 1,604 | 923 | 651 | 407 | 1,981 |
| Proposed Consensus[a] | 308 | 494 | 338 | 1,140 | 574 | 439 | 380 | 1,394 |
| **EMEA** | | | | | | | | |
| Sales | 2,333 | 3,500 | 2,500 | 8,333 | 4,833 | 2,833 | 4,333 | 12,000 |
| Top Down | 2,563 | 2,830 | 2,987 | 8,380 | 3,267 | 2,333 | 2,801 | 8,401 |
| Statistical | 1,167 | 2,177 | 3,444 | 6,788 | 3,542 | 2,731 | 1,919 | 8,193 |
| Proposed Consensus[a] | 2,448 | 3,165 | 2,743 | 8,357 | 3,881 | 2,633 | 3,018 | 9,531 |
| **APAC** | | | | | | | | |
| Sales | 1,217 | 993 | 968 | 3,178 | 1,260 | 880 | 667 | 2,807 |
| Top Down | 366 | 404 | 427 | 1,197 | 467 | 333 | 400 | 1,200 |
| Statistical | 315 | 220 | 595 | 1,129 | 490 | 451 | 398 | 1,339 |
| Proposed Consensus[a] | 791 | 699 | 698 | 2,188 | 739 | 555 | 488 | 1,782 |

Source: Company data.

[a]Proposed consensus shown in Q2 is average of Sales and Top's Down in Q3; it is the average of all three forecasts.

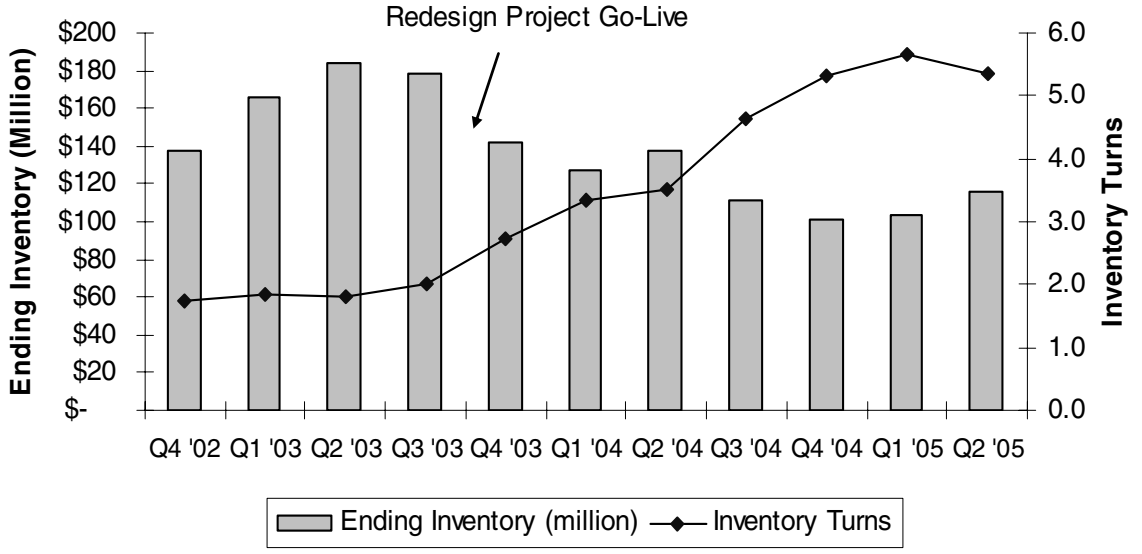
**Exhibit 7** Leitax's Family-Level, Three-Month-Ahead Forecast Accuracy Performance




Source: Company data.

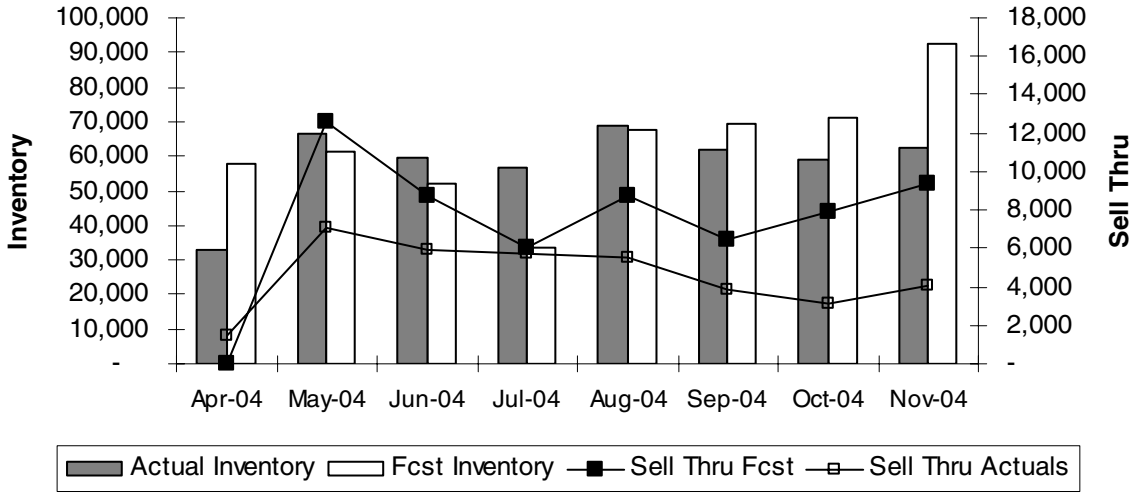
Note: The family-level forecast accuracy is a weighted average by percentage sales (for each product) of the forecast accuracy for each product category.

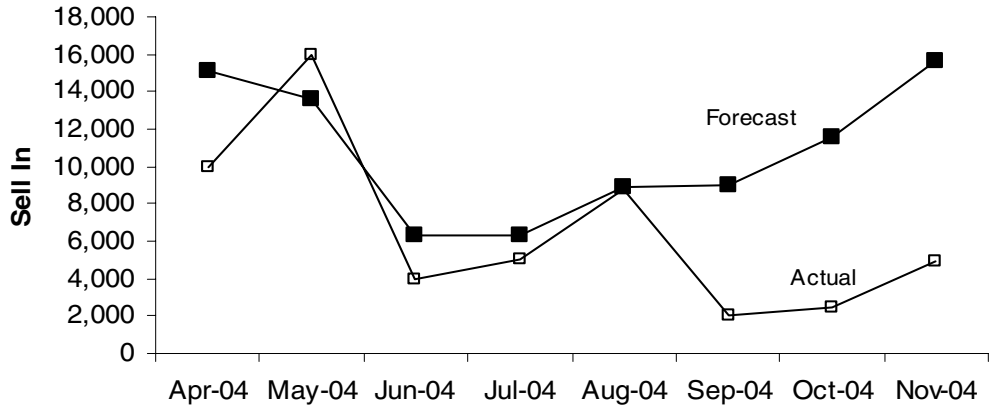
**Exhibit 8** Leitax's Inventory Performance




Source: Company data.

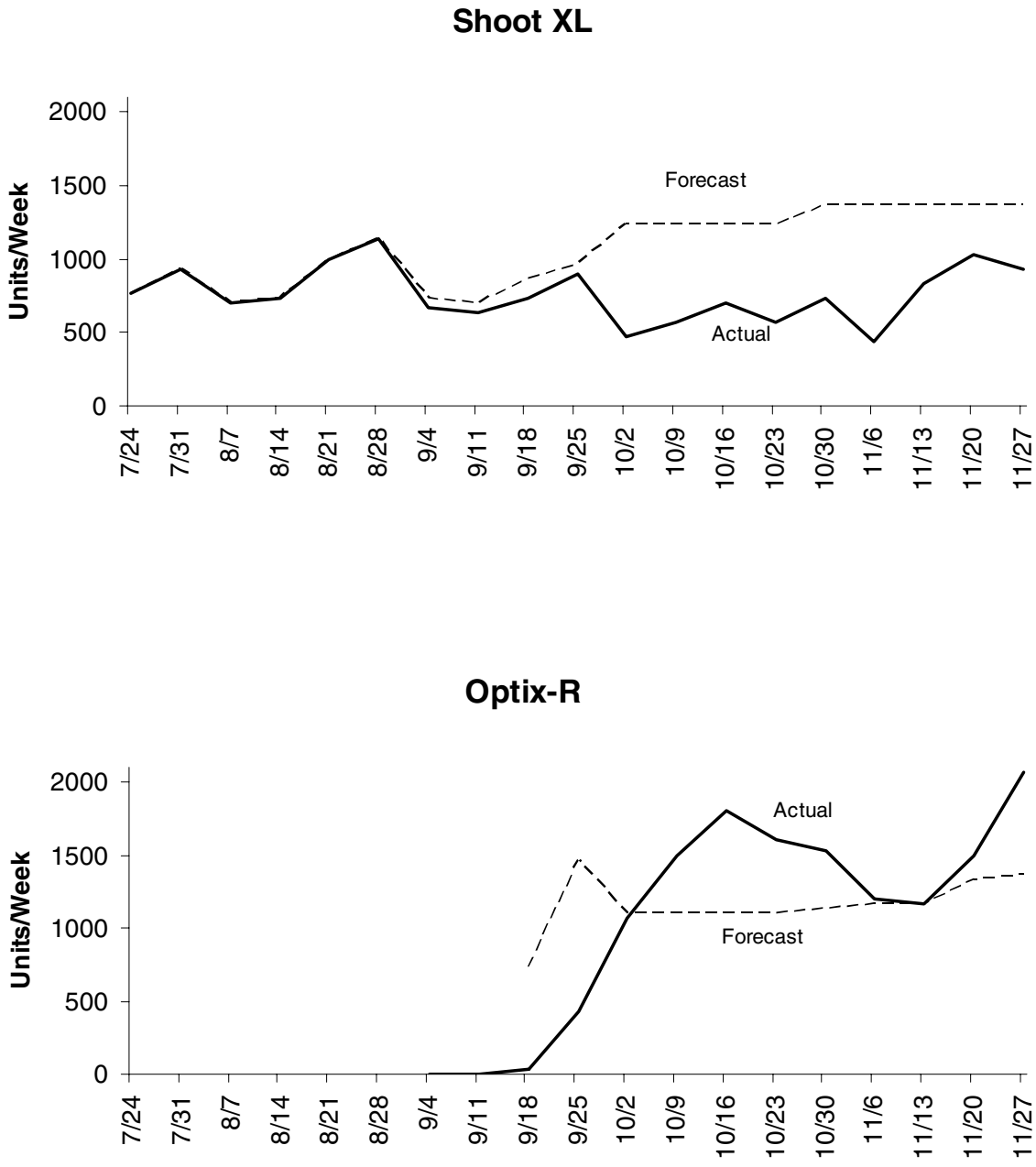
**Exhibit 9** SF 6000 Sales, Worldwide Three-Month-Ahead Forecasts and Actual Channel Inventory Levels




Source: Company data.

**Exhibit 10** Shoot XL 2004 and Optix-R 2004 Worldwide Sales and Three-Month-Ahead Forecast




Source: Company data.